MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

COMPARATOR, ANALOG, KEY CODE GEE

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic comparator module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.930 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be GEE.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 20 contact pins positioned A21 through A40. Contact pins A1 through A20 are optional.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/231-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

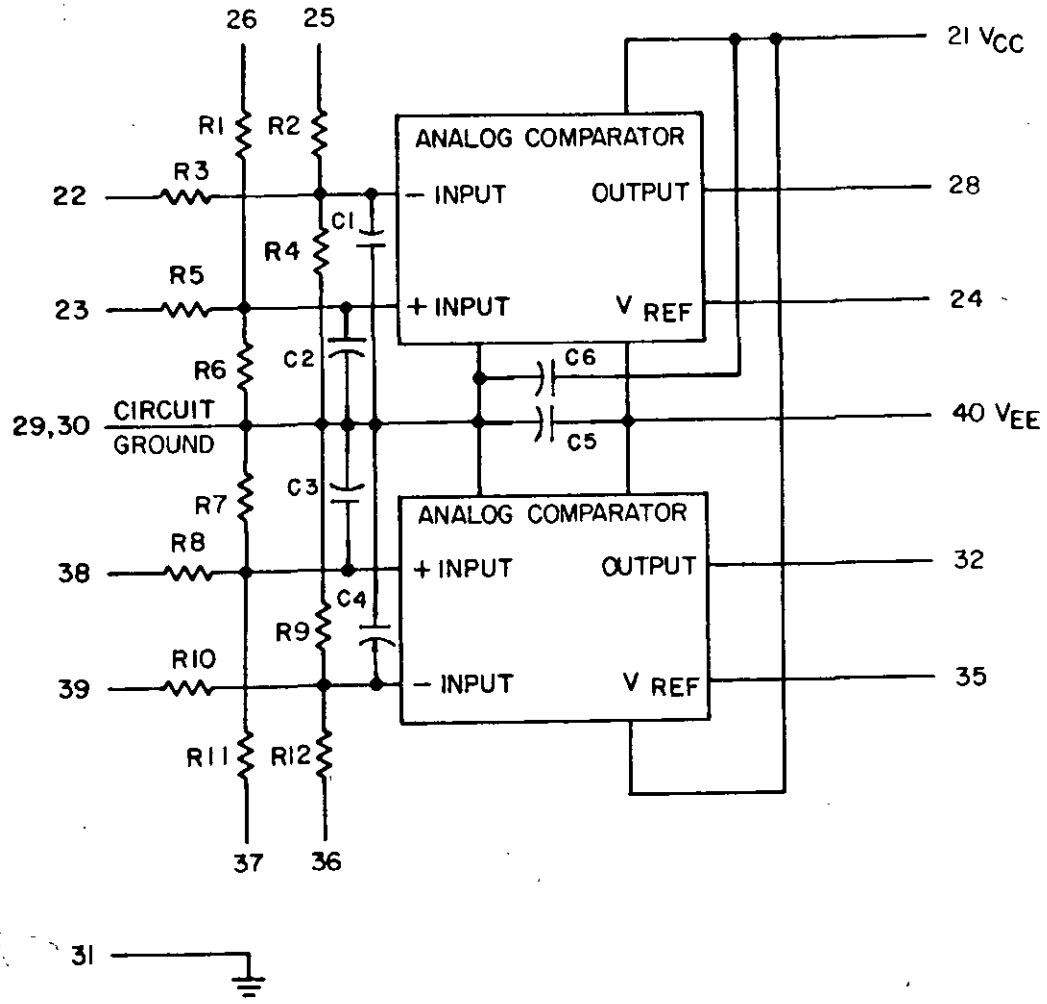
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTES:
1. Pins 27, 33, and 34 are not used. Pin 1 thru 20, if supplied, are not used.
2. V_{REF} designates output low state reference terminal.
3. R1 = R2 = R11 = R12 = 100 kΩ ±1%.
4. R3 = R5 = R8 = R10 = 1.5 kΩ ±1%.
5. R4 = R6 = R7 = R9 = 10 kΩ ±1%.
6. C1-C4 = 100 pF ±10%.
7. C5 = C6 = 0.01 µF ±10%.

FIGURE 1. GEE module function diagram, analog comparator.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1,A2	Isolation	II	S	10	∞	1	∞	10	∞	1	∞	MΩ
B	Stress	III	S	1	2	1	2	1	2	1	2	Minute
C1,C2	Continuity	IV	S	0	1	0	1	0	1	0	1	Ω
D1,D2	V_{OUT} (low loaded)	V	X	0	400	0	420	0	420	0	440	mV
E1,E3	Input offset voltage, low level	VI	X	0	400	0	420	0	420	0	440	mV
E2,E4	Input offset voltage, high level	VI	X	4.60	5.05	4.60	5.05	4.60	5.05	4.60	5.05	V
F1,F3	Output voltage, high level	VII	X	2.4	4.55	2.4	4.55	2.4	4.55	2.4	4.55	V
F2,F4	Output voltage, low level	VII	X	0	400	0	420	0	420	0	440	mV
G1,G3	Response time, high level, 5 V	VIII	S	0	1,000	0	1,000	0	1,000	0	1,000	ns
G2,G4	Response time, low level, 5 V	VIII	S	0	500	0	500	0	500	0	500	ns
H1,H3	Response time, high level, ±15 V	IX	Q	0	1,800	0	1,800	0	1,800	0	1,800	ns
H2,H4	Response time, low level, ±15 V	IX	Q	0	500	0	500	0	500	0	500	ns
J1,J2	Output crosstalk, V_{CC} = 5 V	X	Q	0	400	0	420	0	420	0	440	mV
K1,K3, K6,K8	Input network	XI	S	-14.50	-15.15	-14.50	-15.15	-14.50	-15.15	-14.50	-15.15	V
K2,K4, K5,K7	Input network	XI	S	14.50	15.15	14.50	15.15	14.50	15.15	14.50	15.15	V
L1-L4	Input resistance, high level	XII	X	108	112	108	112	105	115	105	115	kΩ
L5-L8	Input resistance, low level	XII	X	11.3	11.7	11.3	11.7	11	12	11	12	kΩ
L9-L12	Input resistance, low level to high level	XII	X	99.5	103.5	99.5	103.5	96.5	106.5	96.5	106.5	kΩ
M1	Positive supply current, high level	XIII	Q	2	18	2	20	2	22	2	24	mA

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
M2	Negative supply current, high level	XIII	Q	2	15	2	17	2	19	2	21	mA
M3	Positive supply current, low level	XIII	Q	15	33	15	35	15	37	15	39	mA
M4	Negative supply current, low level	XIII	Q	15	30	15	32	15	34	15	36	mA
N1,N2	Output saturation voltage	XIV	S	0.0	1.6	0.0	1.6	0.0	1.6	0.0	1.6	V
P1,P2	Power supply transient rise time	XV	S	2.5	7.5	2.5	7.5	2.5	7.5	2.5	7.5	µs

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through P shall be performed in any order with the following exceptions. If test conditions A or B are part of a sequence of tests, the order of testing shall be as follows: A, B where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through P in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through P shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Items X01-1 and X01-2 shall be monitored during the test and the readings shall not increase more than 10 percent from the initial current readings taken at maximum class temperatures at the start of the test.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition P, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Mechanical shock, thermal shock, humidity, salt fog, and durability tests. The environmental tests shall be conducted in accordance with MIL-M-28787. Following these tests, the electrical tests of table I marked with an (X) in the test type column shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.4 Life test. The life test shall be conducted in accordance with MIL-M-28787 except the module shall be operated for a minimum of 1000 hours. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through P shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through P shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Military specification MIL-M-28787/231 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2833711. Part number M28787/231-1 supersedes part numbers 2833711 and M28787/231.

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6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Maximum power dissipation (V_{CC} = +16.5 V, V_{EE} = -16.5 V, V_{REF} = -16.5 V, low level)		1.24 W (outputs not loaded)
Supply voltages		V_{CC} = +5 to +15 V dc ±10% V_{EE} = -5 to -15 V dc ±10% V_{REF} = 0 to V_{EE}
Input voltage		
Pins 25, 26, 36, 37		V_{CC} x 10 maximum V_{EE} x 10 maximum
Pins 22, 23, 38, 39		V_{CC} maximum V_{EE} maximum
Input offset voltage		
Pin 22 to pin 23		±6 mV maximum
Pin 38 to pin 39		±6 mV maximum
Input resistance		
Pins 25, 26, 36, 37		110 kΩ ±5%
Pins 22, 23, 38, 39		11.5 kΩ ±5%
Output voltage		
Low level		0.44 V maximum
High level		2.4 V minimum
Output sink current, low level		6.4 mA minimum
Output source current, high level		400 µA minimum
Output saturation voltage (Output sink current = 40 mA)		1.6 V dc maximum
Turn-on response time	V_{CC} = 5 V	1 µs maximum
	V_{CC} = 15 V	1.8 µs maximum
Turn-off response time		500 ns maximum
Quiescent power supply current		
V_{CC}		39 mA maximum
V_{EE} (V_{REF} = V_{EE})		36 mA maximum

6.5 Module function. This module contains two identical voltage comparators with independent inputs and outputs as illustrated on figure 1. Both normal and high level inverting and noninverting inputs are provided. An independent low-state level reference terminal is provided for each comparator output. Output terminal has a 4.7 kΩ ±5 percent source capability. The module meets the logic output requirements of 1.2 V for a V_{CC} supply voltage of 4 V ±10 percent with V_{REF} terminal at 0 V.

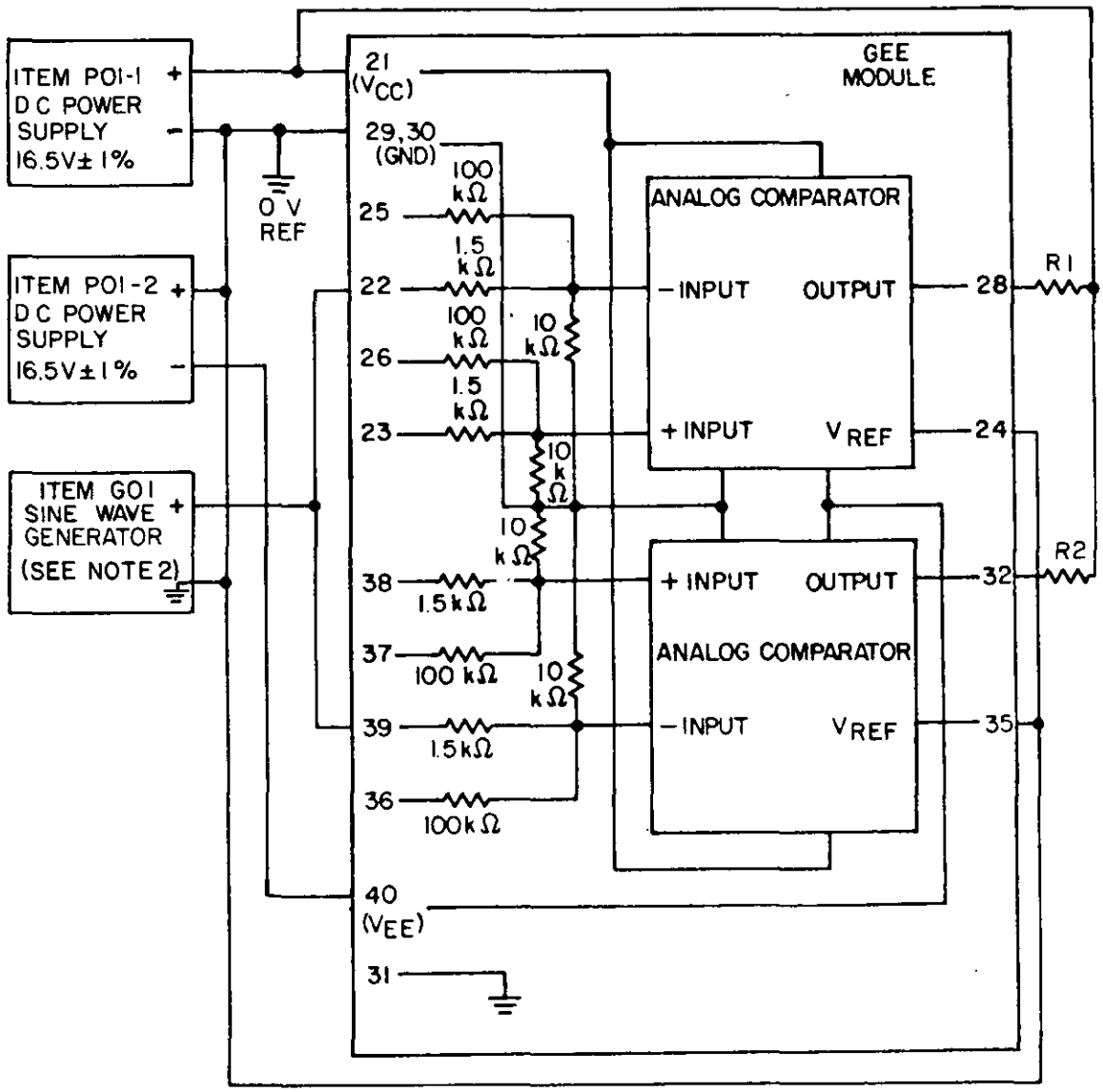
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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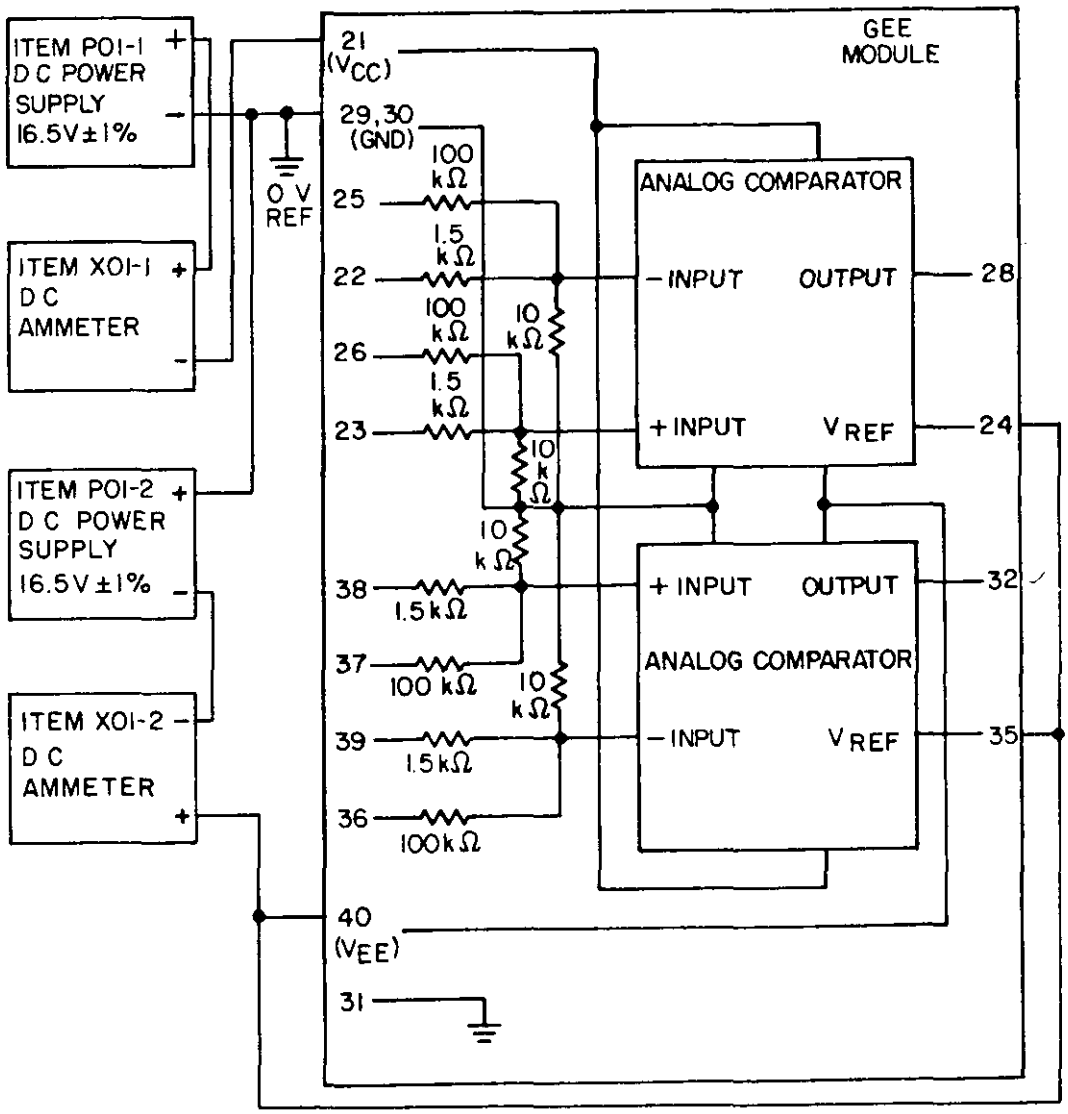
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NOTES:
1. R1 = R2 = 825Ω ±1%, 2W.
2. Item G01 setup:
   Amplitude = 2 V rms ±1%.
   Frequency = 10 kHz ±1%.

*FIGURE 2. Operating temperature and life test circuit.

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CAUTION: Do not isolate the chassis of item XO1 from earth ground.

FIGURE 3. Component temperature test circuit.

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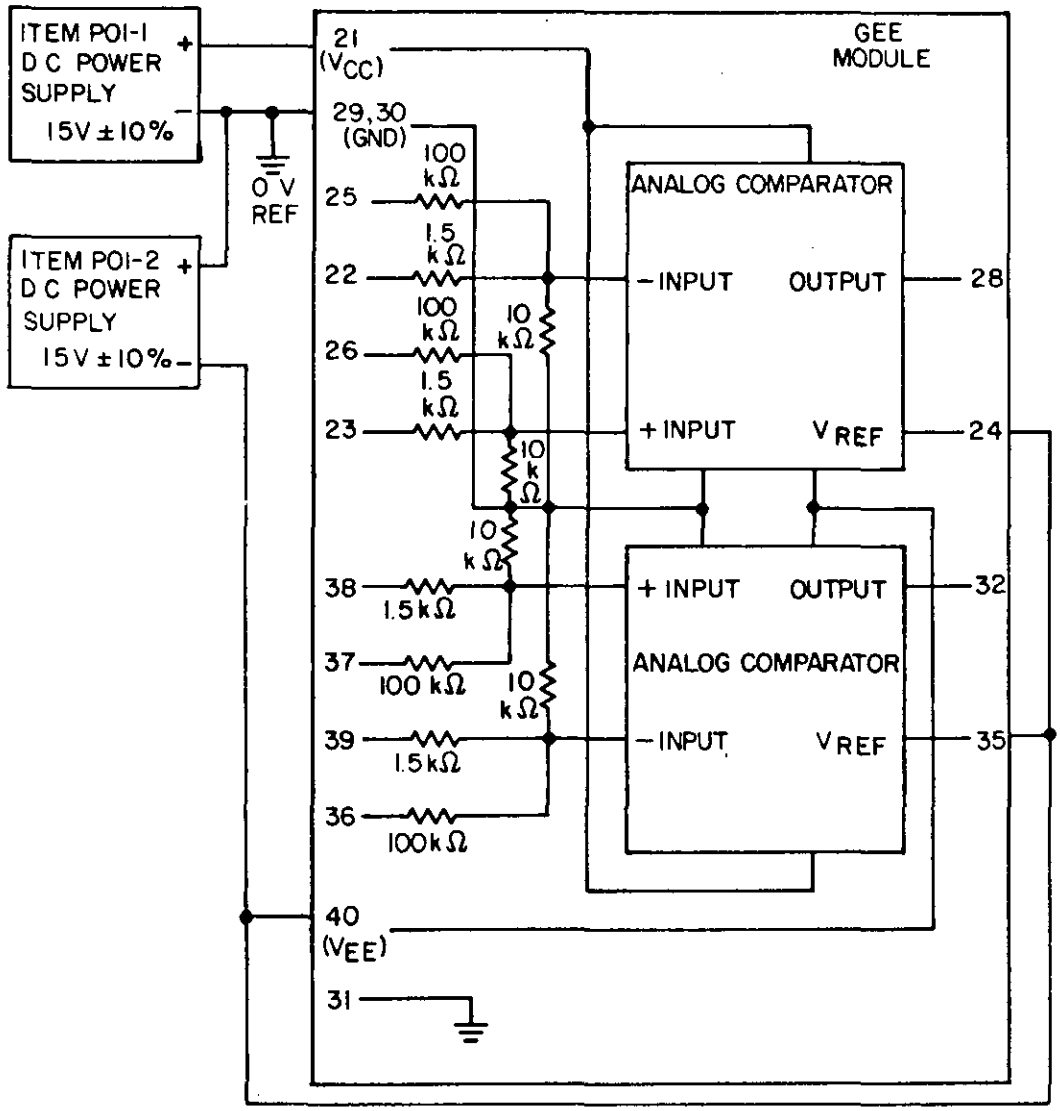


*FIGURE 4. Vibration test circuit.

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TABLE II. Isolation test.

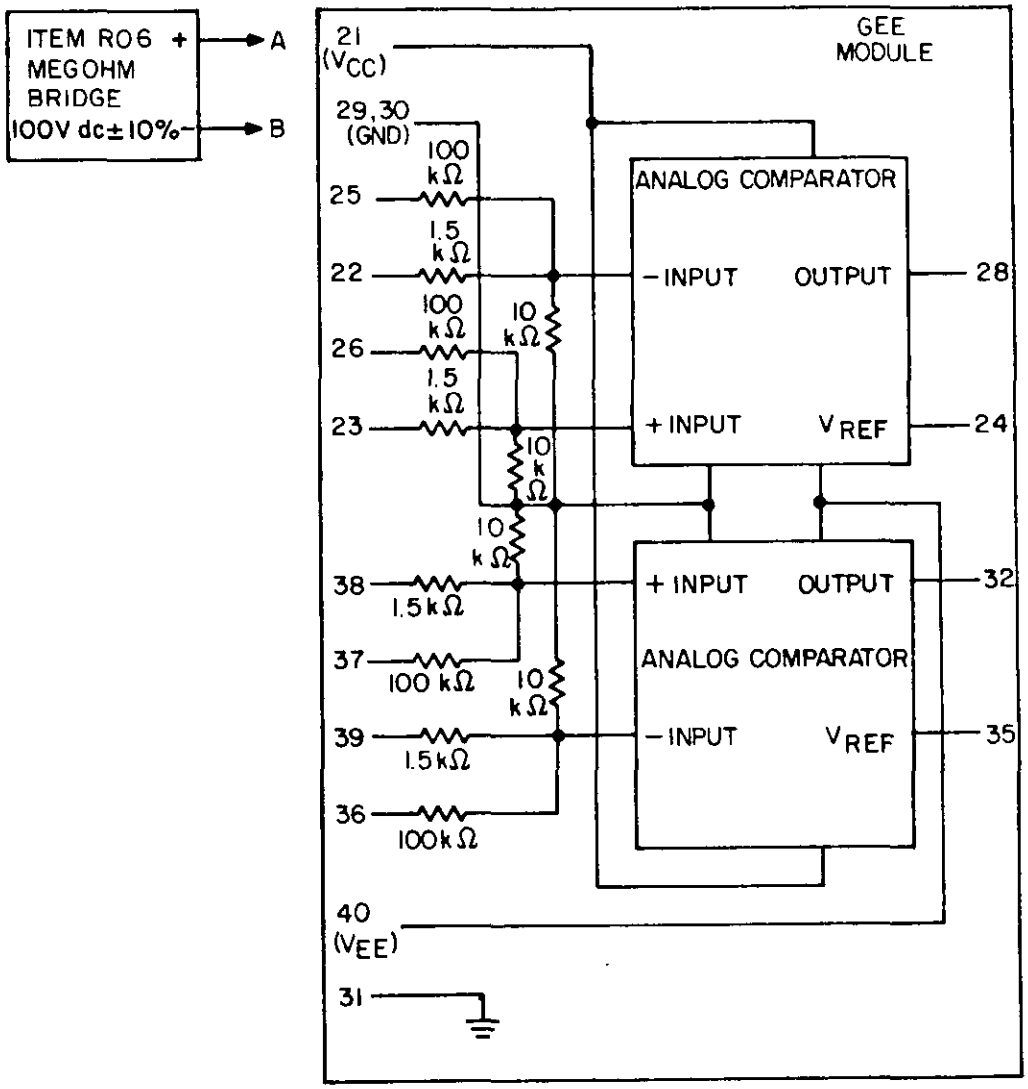
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	All other pins	31	10	∞	MΩ	5
A2	Isolation, unused pin 1/	All other pins	27, 33, 34	10	∞	MΩ	5

1/ If the optional 20 pins are supplied, they shall be included under the "B to pin" for test condition A2.

TABLE III. Stress test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
B	Maintain stress condition for 1 to 2 minutes. Module shall pass remainder of this specification as proof of no damage.	1	2	Minute	6

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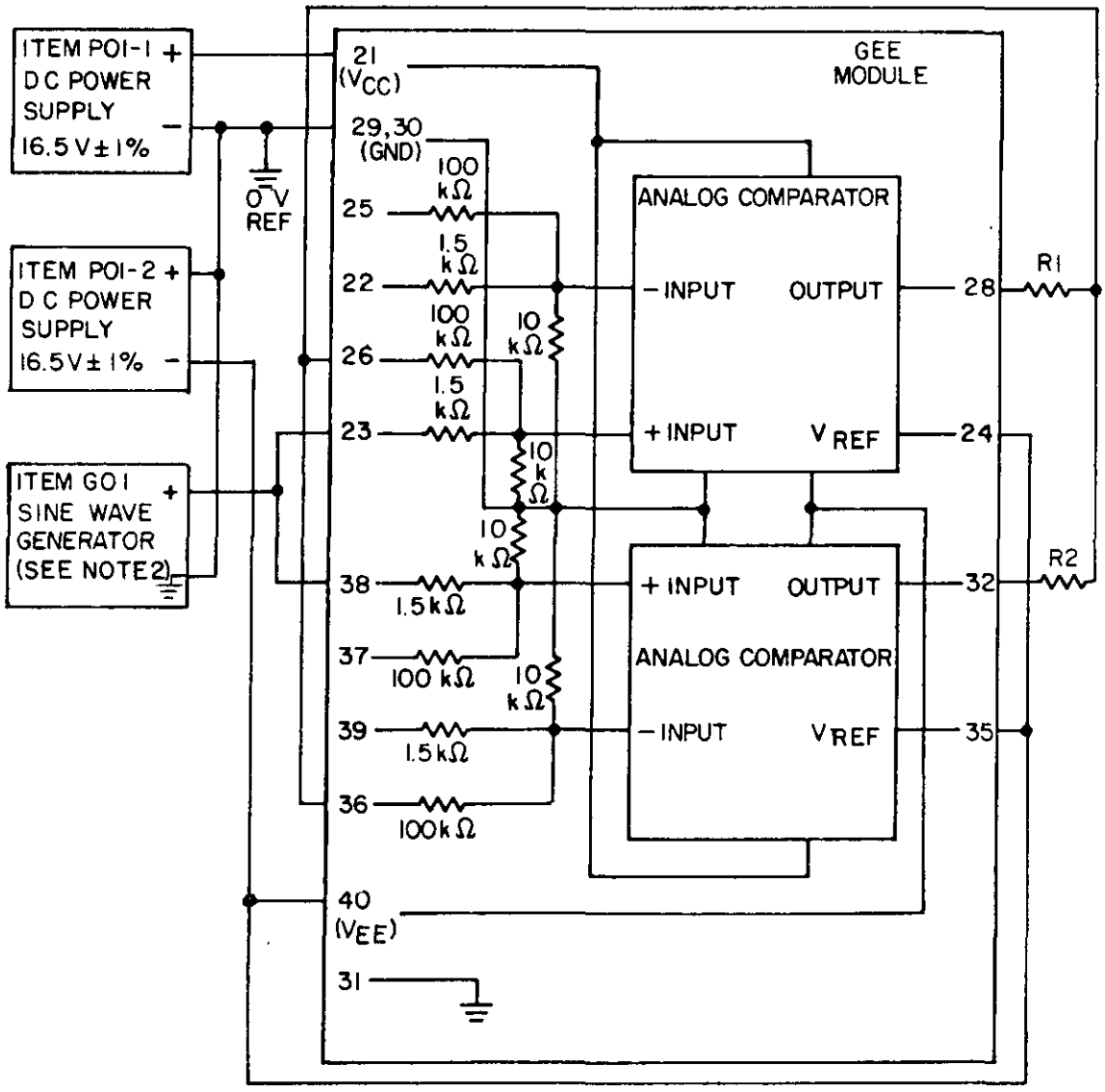


WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

FIGURE 5. Isolation test circuit for table II.

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NOTES:
1. R1 = R2 = 825Ω ±1%, 2 W.
2. Item G01 setup:
   Amplitude = 10 V rms ±1%.
   Frequency = 10 kHz ±10%.

*FIGURE 6. Stress test circuit for table III.

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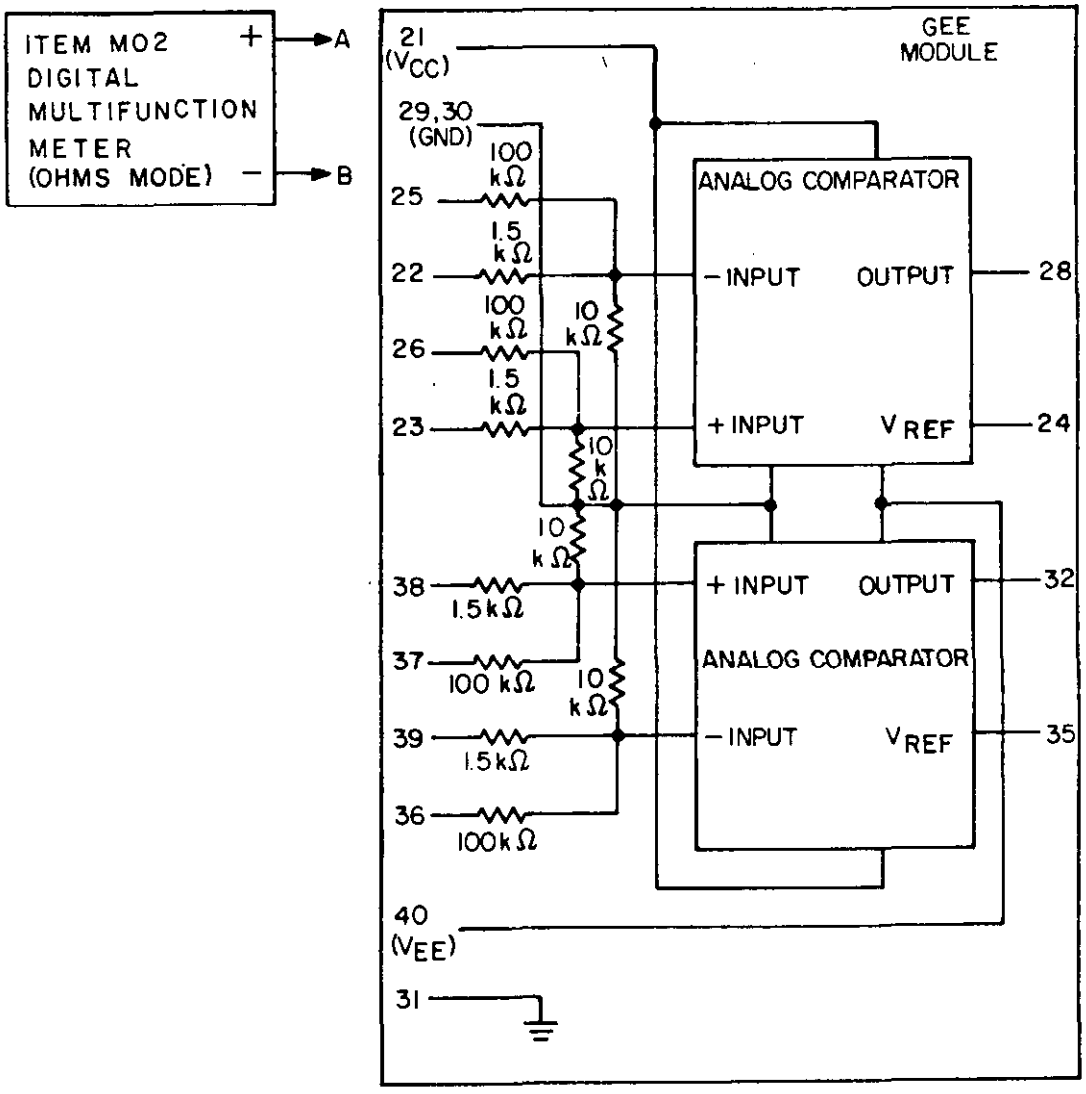
TABLE IV. Continuity test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Continuity	Frame	31	0	1	Ω	7
C2	Continuity	29	30	0	1	Ω	7

TABLE V. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
D1	V_{OUT} (low loaded)	28	0	400	mV	8
D2	V_{OUT} (low loaded)	32	0	400	mV	8

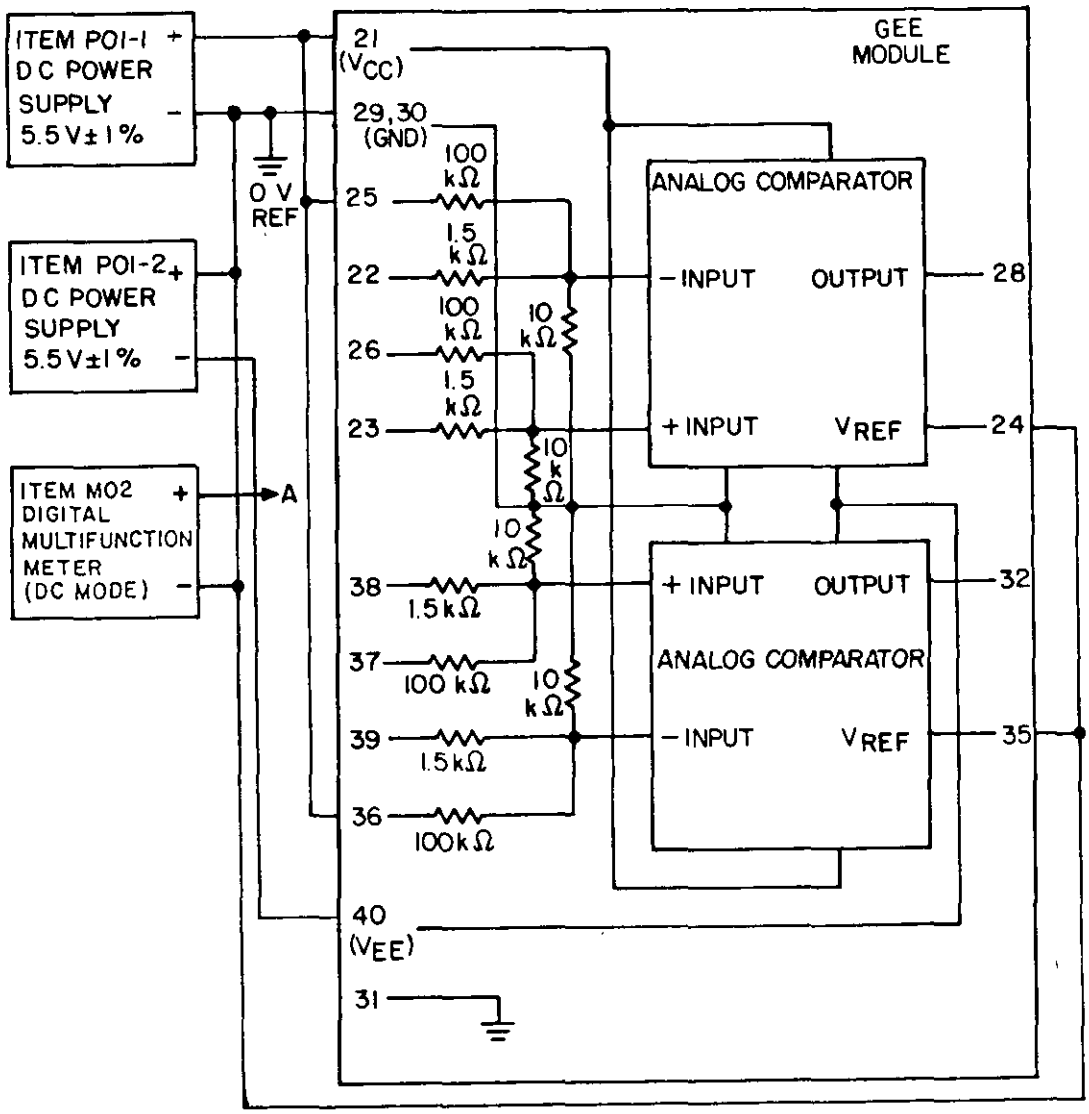
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NOTE: For test condition C1, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the positive terminal.

*FIGURE 7. Continuity test circuit for table IV.

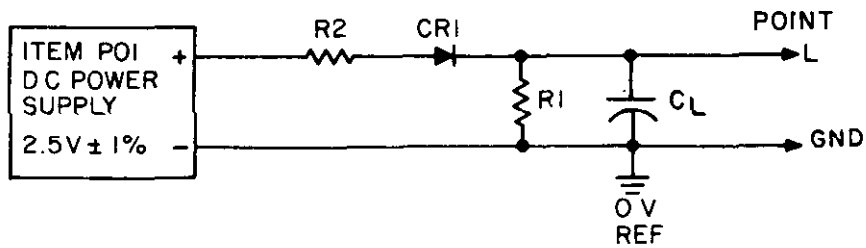
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NOTE: Load output with a load as shown on figure 9.

*FIGURE 8. V_{OUT} (low-loaded) test circuit for table V.

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NOTES:
1. CR1 = 1N4148, or equivalent.
2. C_L = 39 ±5 pF (including probe and parasitic capacitance).
3. R1 = 6.04 kΩ ±1%, 1/8 W.
4. R2 = Select value for current flow of 6.4 mA out of point L with point L held to 0.4 V.

*FIGURE 9. GEE logic load circuit.

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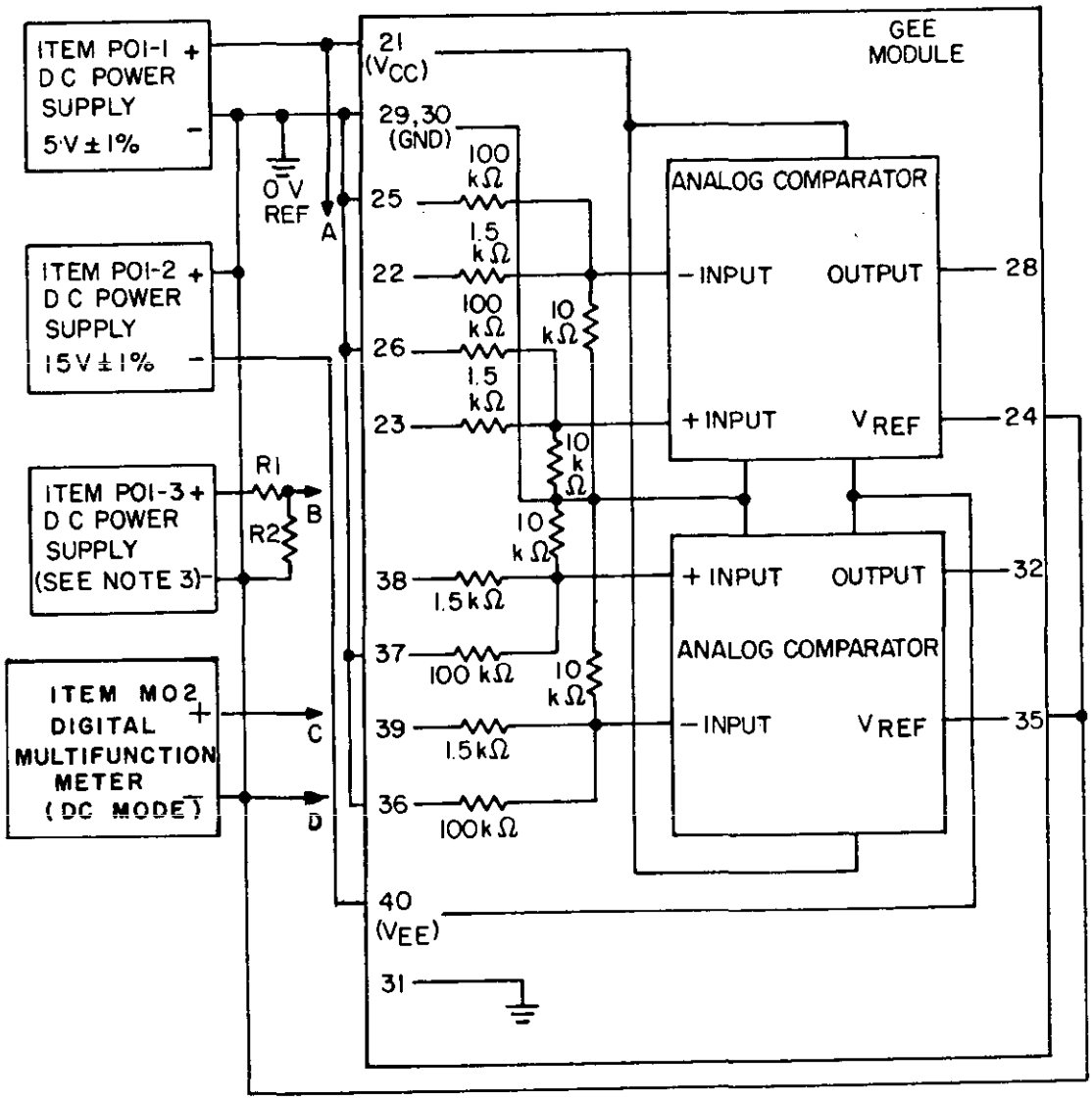
TABLE VI. Input offset voltage (low-level and high-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
E1	Input offset voltage, low level	39	22	28	23,38	0	400	mV	10
E2	Input offset voltage, high level	39	23	28	22,38	4.60	5.05	V	"
E3	Input offset voltage, low level	22	39	32	23,38	0	400	mV	"
E4	Input offset voltage, high level	22	38	32	23,39	4.60	5.05	V	"

*TABLE VII. Output voltage (high-level and low-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
F1	Output voltage, high level	23	28	2.40	4.55	V	11
F2	Output voltage, low level	22	28	0	400	mV	"
F3	Output voltage, high level	28	32	2.40	4.55	V	"
F4	Output voltage, low level	39	32	0	400	mV	"

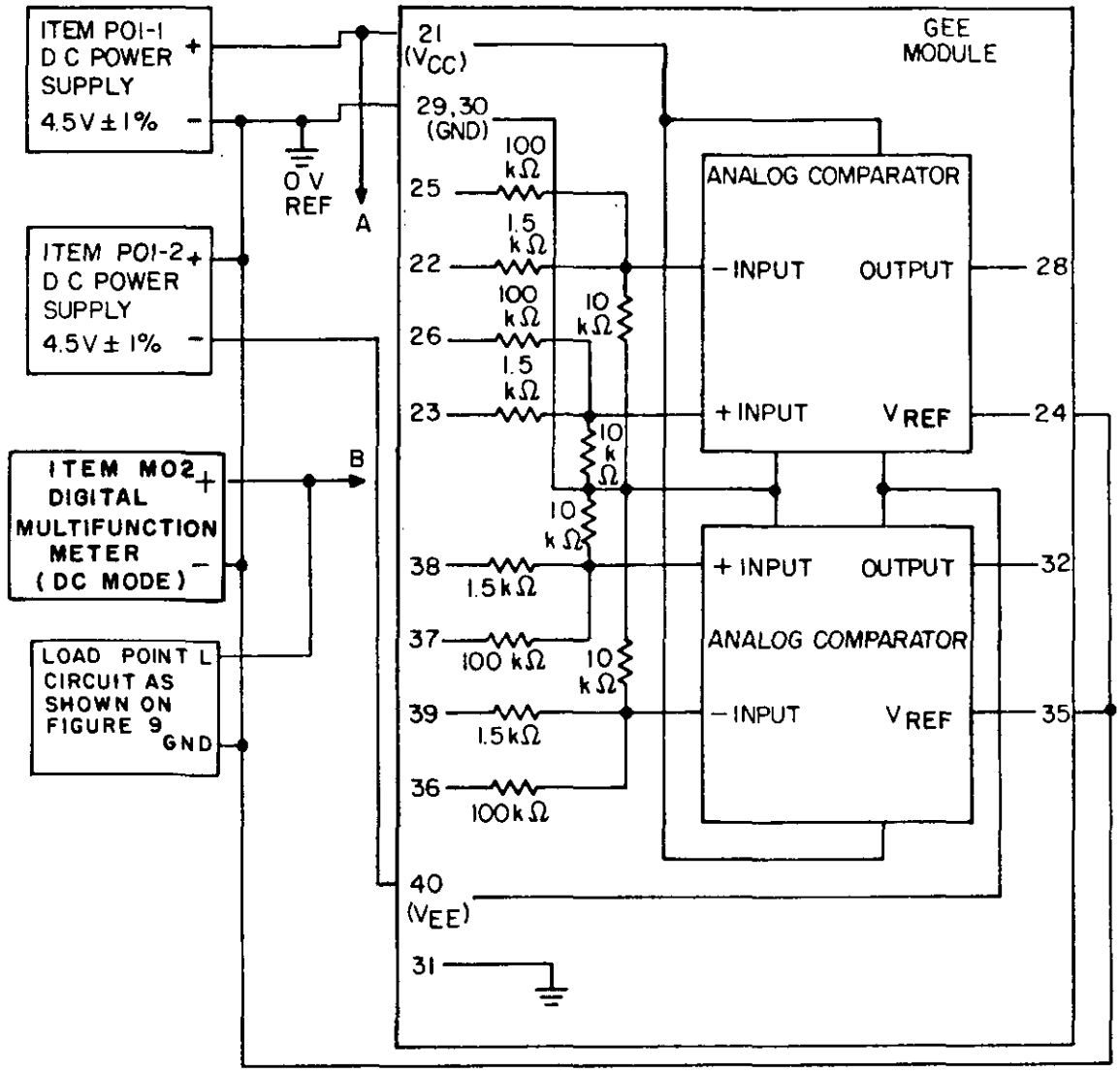
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NOTES:
1. R1 = 4530Ω ±1%, 1/8 W.
2. R2 = 49.9Ω ±1%, 1/8 W.
3. Adjust item PO1-3 to provide 6 mV ±1% at point B with point B connected.

*FIGURE 10. Input offset voltage (low-level and high-level) test circuit for table VI.

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* FIGURE 11. Output voltage (high-level and low-level) test circuit for table VII.

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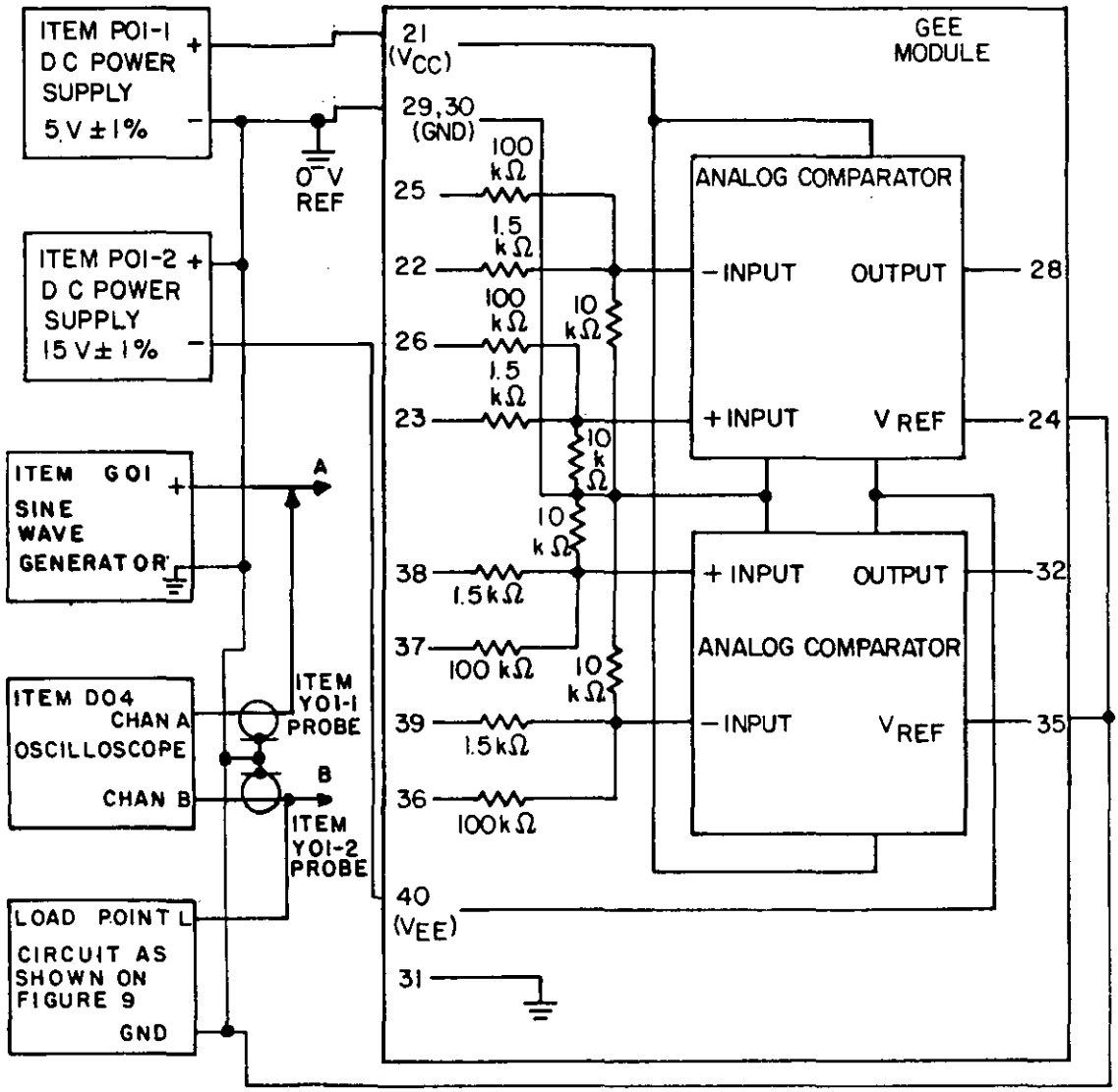
TABLE VIII. Response time (high-level and low-level, 5 V supply) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
G1	Response time, high level, 5 V	23	28	0	1,000	ns	12,13
G2	Response time, low level, 5 V	23	28	"	500	"	"
G3	Response time, high level, 5 V	38	32	"	1,000	"	"
G4	Response time, low level, 5 V	38	32	"	500	"	"

TABLE IX. Response time (high-level and low-level, ±15 V supply) test.

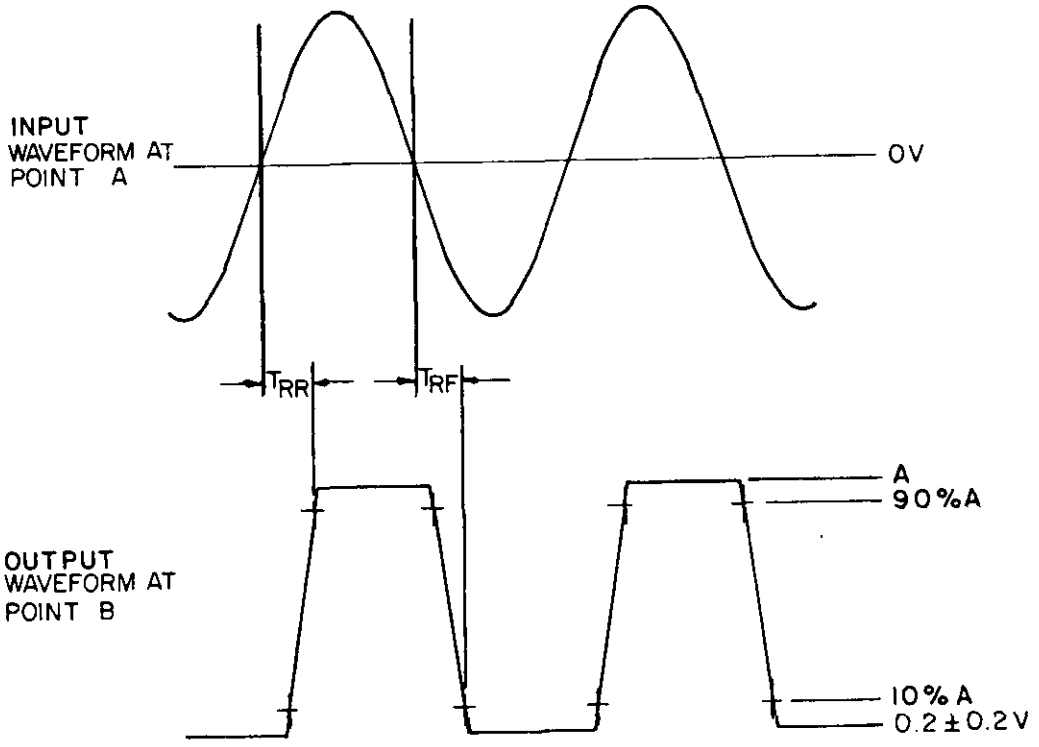
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H1	Response time, high level, ±15 V	23	28	0	1,800	ns	14,15
H2	Response time, low level, ±15 V	23	28	"	500	"	"
H3	Response time, high level, ±15 V	38	32	"	1,800	"	"
H4	Response time, low level, ±15 V	38	32	"	500	"	"

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* FIGURE 12. Response time (high-level and low-level, 5 V supply) test circuit for table VIII.

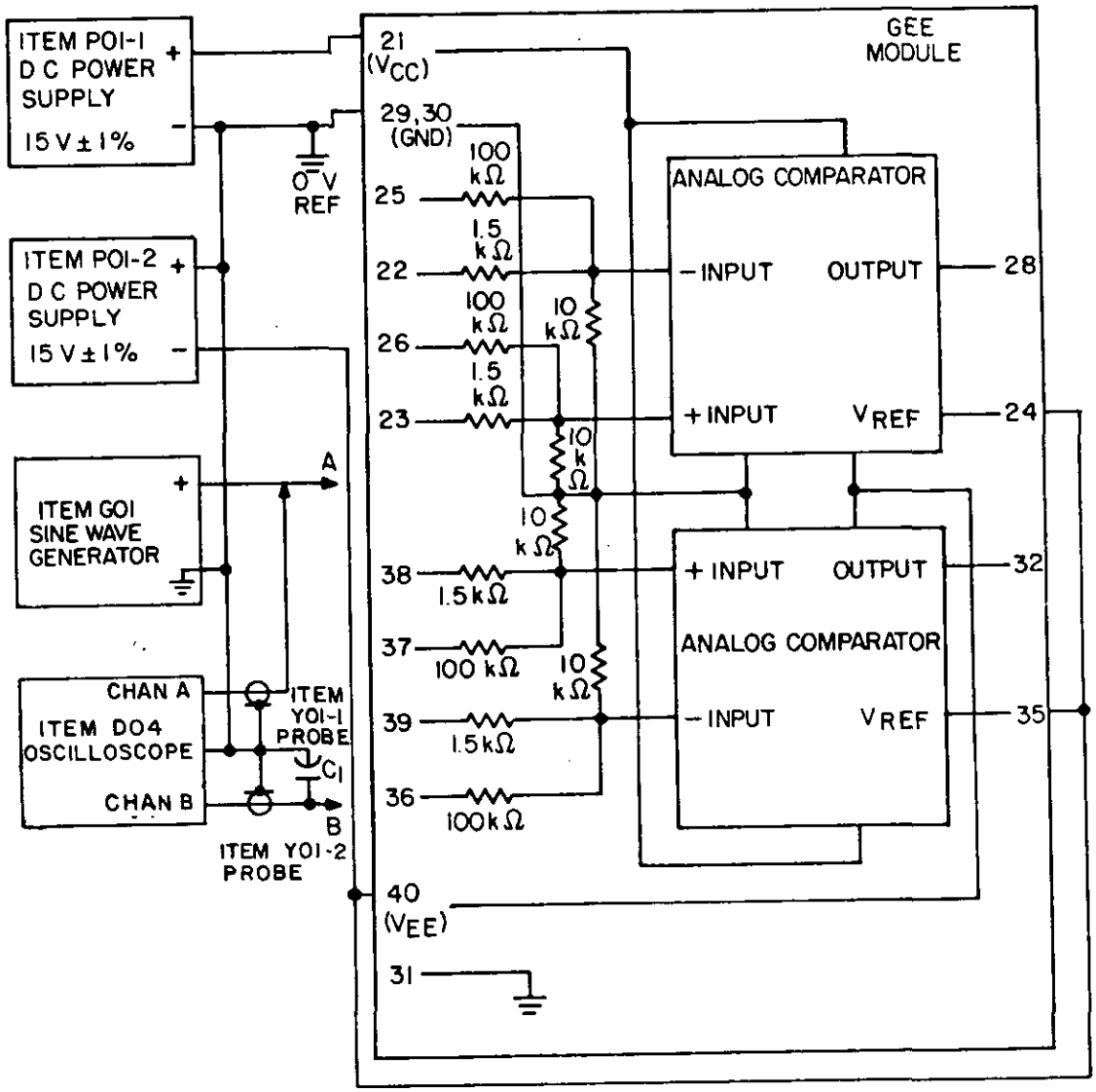
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NOTES:
1. Input waveform parameters: f = 100 kHz ±1% at 1 V rms ±1%.
2. Test condition G1 and G3; measure T_{RR} at point B.
3. Test condition G2 and G4; measure T_{RF} at point B.

FIGURE 13. Response time (high-level and low-level, 5 V supply) test circuit waveforms for table VIII.

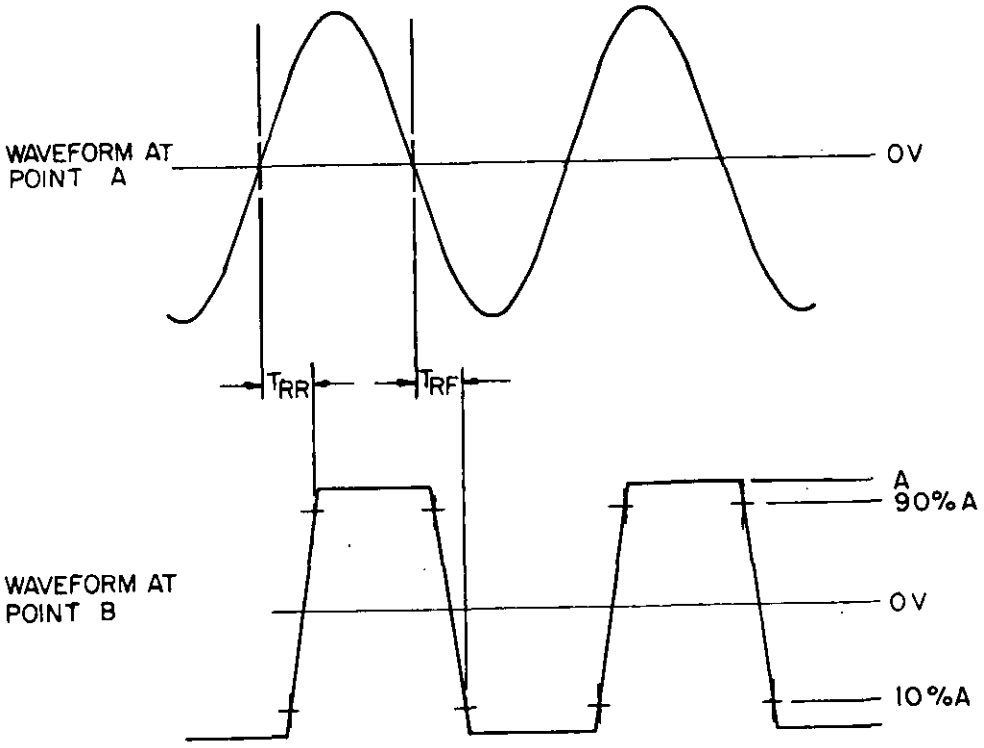
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NOTE: C1 = 50 ±5 pF (including probe and parasitic capacitance).

FIGURE 14. Response time (high-level and low-level, ±15 V supply) test circuit for table IX.

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NOTES:
1. Input waveform parameters: f = 100 kHz ±1% at 1 V rms ±1%.
2. Test condition H1 and H3; measure T_{RR} at point B.
3. Test condition H2 and H4; measure T_{RF} at point B.

FIGURE 15. Response time (high-level and low-level, ±15 V supply) test circuit waveforms for table IX.

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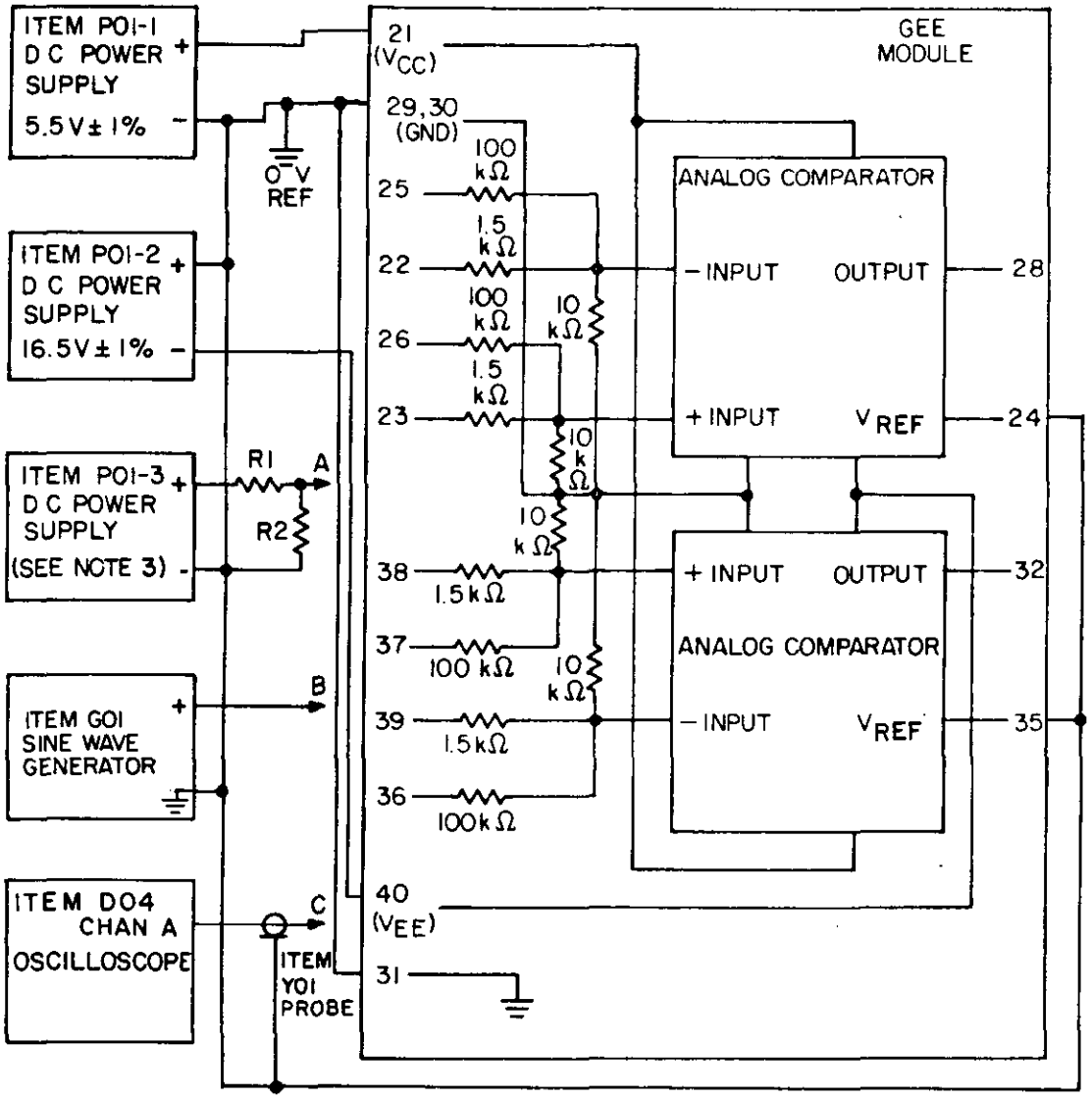
TABLE X. Output crosstalk (V_{CC} = 5 V) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	Output crosstalk, V_{CC} = 5 V	22	39	28	0	400	mV pk	16,17
J2	Output crosstalk, V_{CC} = 5 V	39	22	32	0	400	mV pk	16,17

*TABLE XI. Input network test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
K1	Input network	25,36	23	28	-14.50	-15.15	V	18
K2	"	26,37	22	28	14.50	15.15	"	"
K3	"	25,36	38	32	-14.50	-15.15	"	"
K4	"	26,37	39	32	14.50	15.15	"	"
K5	"	25,36	23	28	14.50	15.15	"	"
K6	"	26,37	22	28	-14.50	-15.15	"	"
K7	"	25,36	38	32	14.50	15.15	"	"
K8	"	26,37	39	32	-14.50	-15.15	"	"

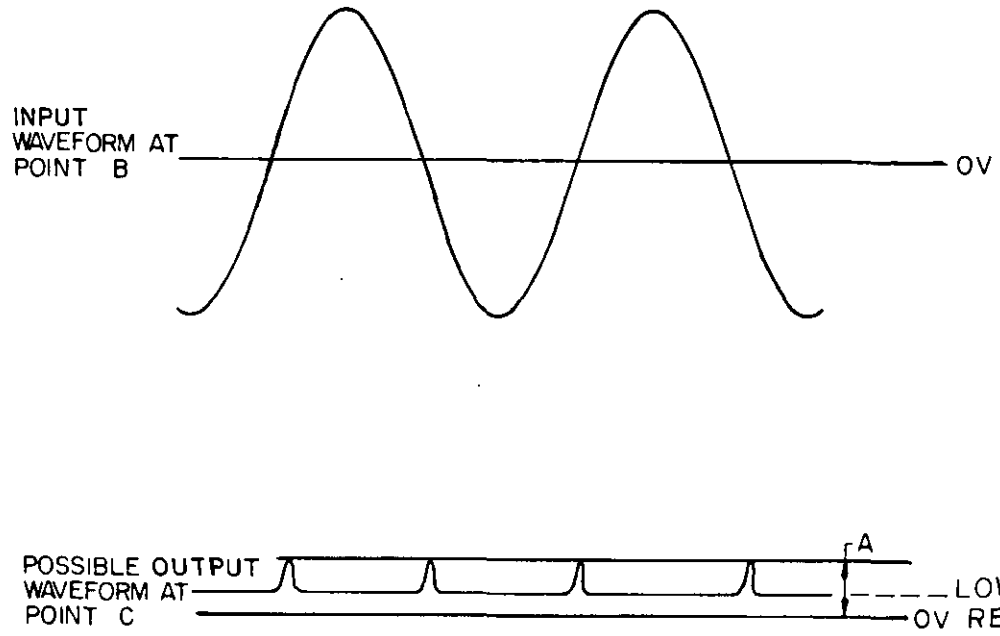
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NOTES:
1. R1 = 4530Ω ±1%, 1/8 W.
2. R2 = 49.9Ω ±1%, 1/8 W.
3. Adjust item P01-3 to provide 20 mV dc ±1% at point A.

FIGURE 16. Output crosstalk (V_{CC} = 5 V) test circuit for table X.

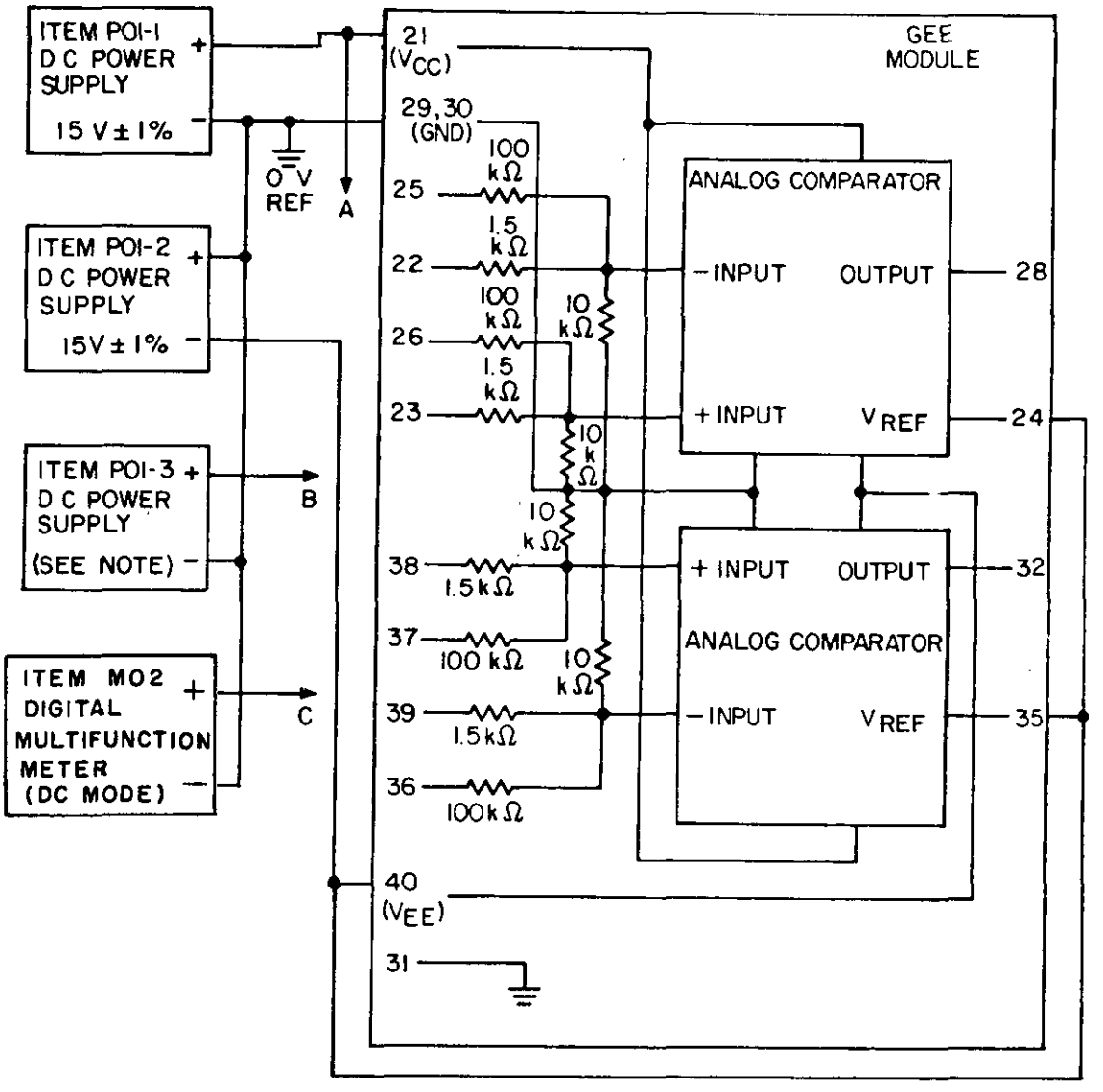
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NOTES:
1. Input waveform parameters: f = 100 kHz ±1% at 1 V rms ±1%.
2. A is parameter under test.

FIGURE 17. *Output crosstalk (V_{CC} = 5 V) test circuit waveforms for table X.*

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NOTE: For test conditions K1-K4 adjust item P01-3 to 1.44 V ±1%.
For test conditions K5-K8 adjust item P01-3 to 1.7 V ±1%.

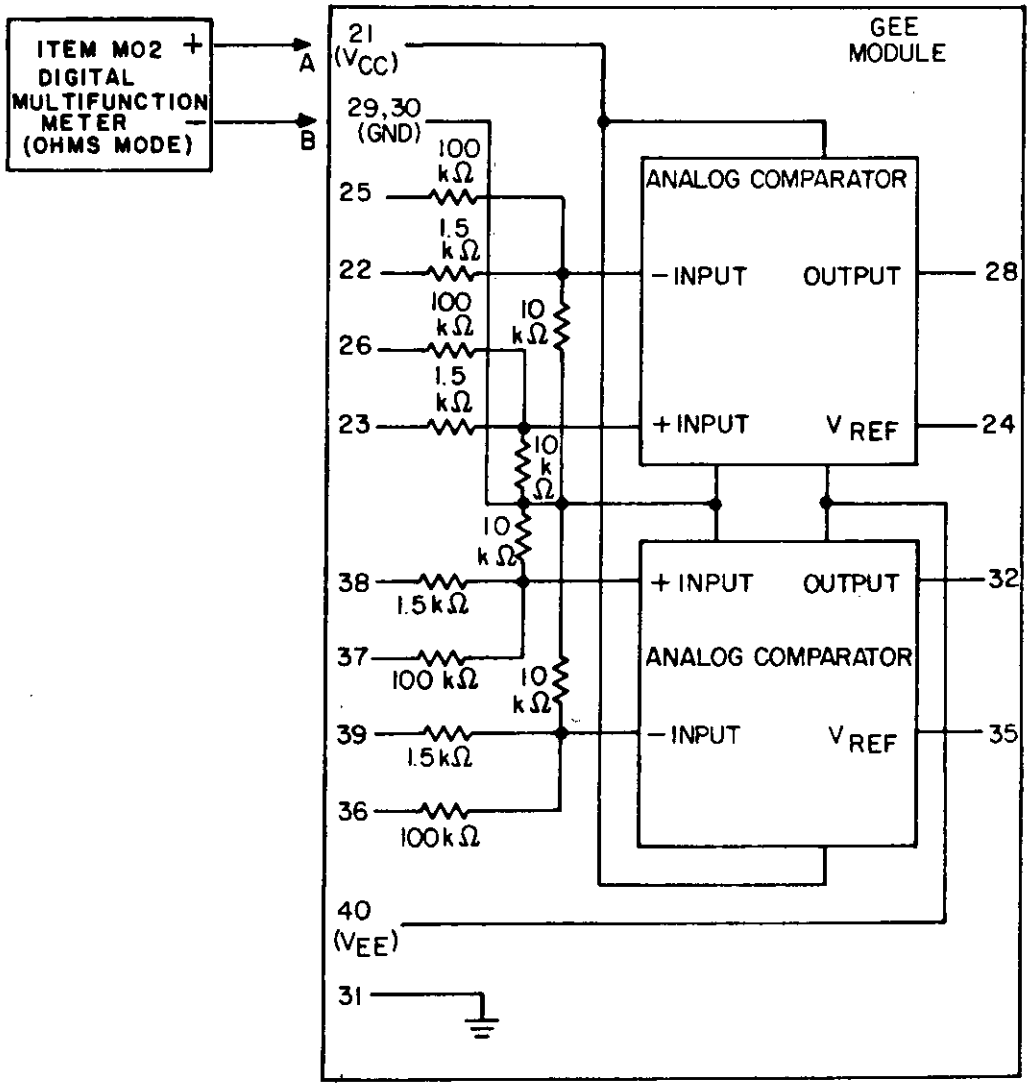
*FIGURE 18. Input network test circuit for table XI.

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TABLE XII. Input resistance (high-level and low-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L1	Input resistance, high level	25	30	108	112	kΩ	19
L2	"	26	"	"	"	"	"
L3	"	36	"	"	"	"	"
L4	"	37	"	"	"	"	"
L5	Input resistance, low level	22	"	11.3	11.7	"	"
L6	"	23	"	"	"	"	"
L7	"	39	"	"	"	"	"
L8	"	38	"	"	"	"	"
L9	Input resistance, low level to high level	22	25	99.5	103.5	"	"
L10	"	23	26	"	"	"	"
L11	"	37	38	"	"	"	"
L12	"	36	39	"	"	"	"

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* FIGURE 19. Input resistance (high-level and low-level) test circuit for table XII.

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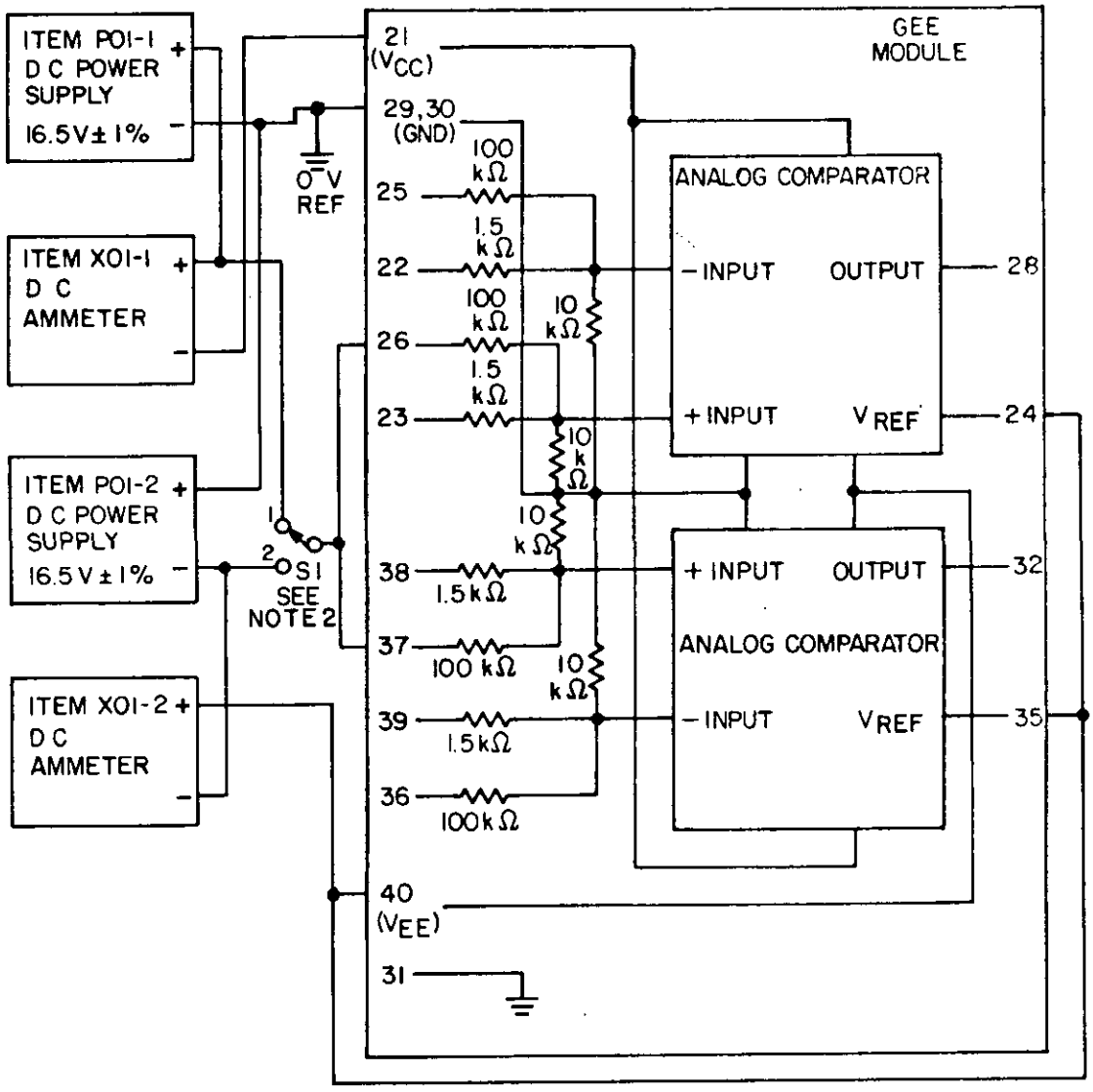
TABLE XIII. Positive and negative power supply current (high-level and low-level) test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
M1	Positive supply current, high level	2	18	mA	20
M2	Negative supply current, high level	2	15	"	"
M3	Positive supply current, low level	15	33	"	"
M4	Negative supply current, low level	15	30	"	"

TABLE XIV. Output saturation voltage test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
N1	Output saturation voltage	28	0.0	1.6	V	21
N2	Output saturation voltage	32	0.0	1.6	V	21

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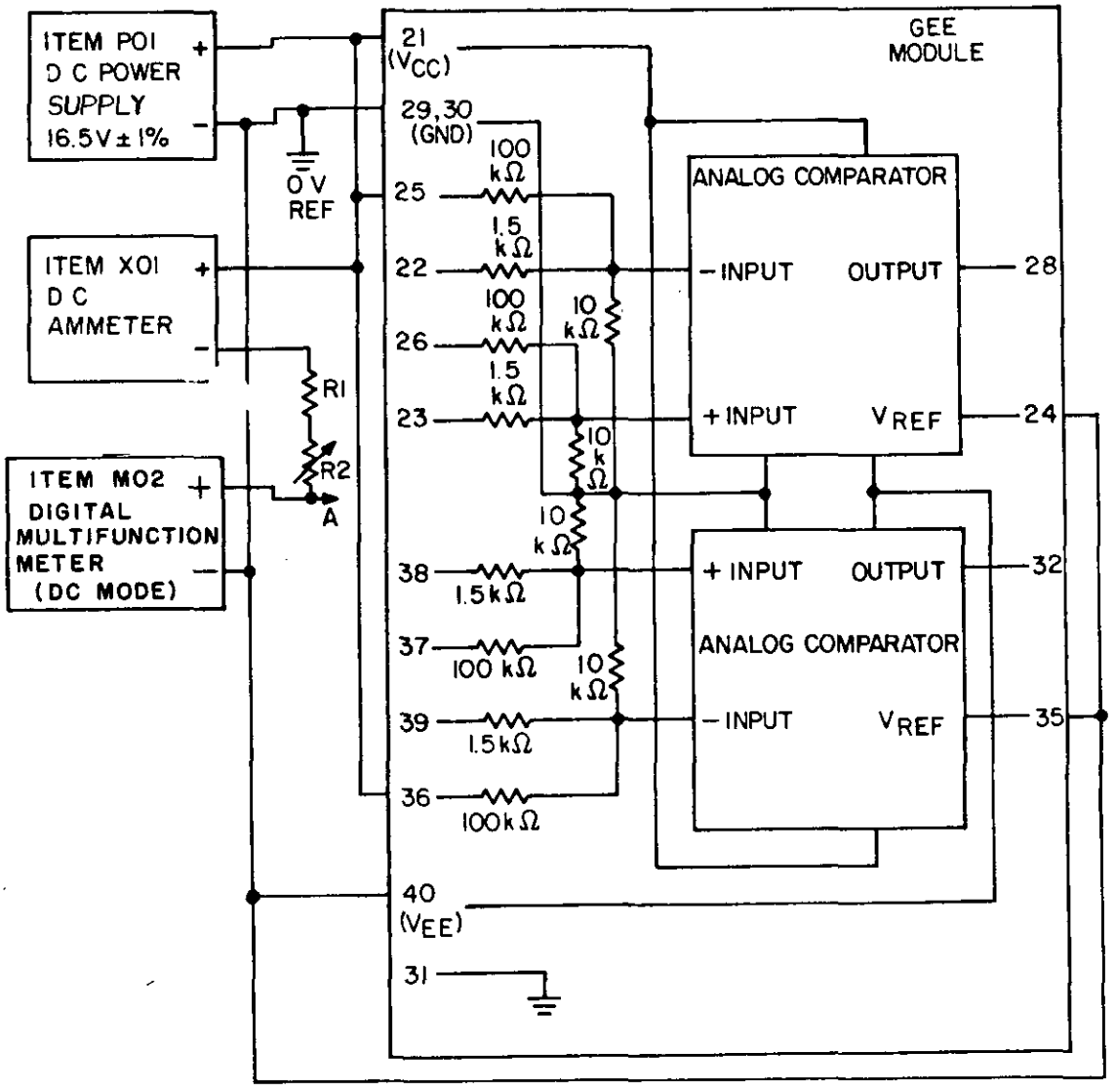


CAUTION: Do not isolate the chassis of item X01 from earth ground.

NOTES:
1. S1 = SPDT switch.
2. For test conditions M1 and M2 set switch S1 to position 1. For test conditions M3 and M4 set switch S1 to position 2.
3. Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required.

FIGURE 20. Positive and negative power supply current (high-level and low-level) test circuit for table XIII.

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CAUTION: Do not isolate the chassis of item X01 from earth ground.

NOTES:
1. R1 = 357Ω ±1%, 1 W.
2. R2 = 100Ω, 1 W, variable resistor.
3. Initially adjust R2 for a current of 40 ±1 mA as read on item X01 with point A connected as specified in table XIV.

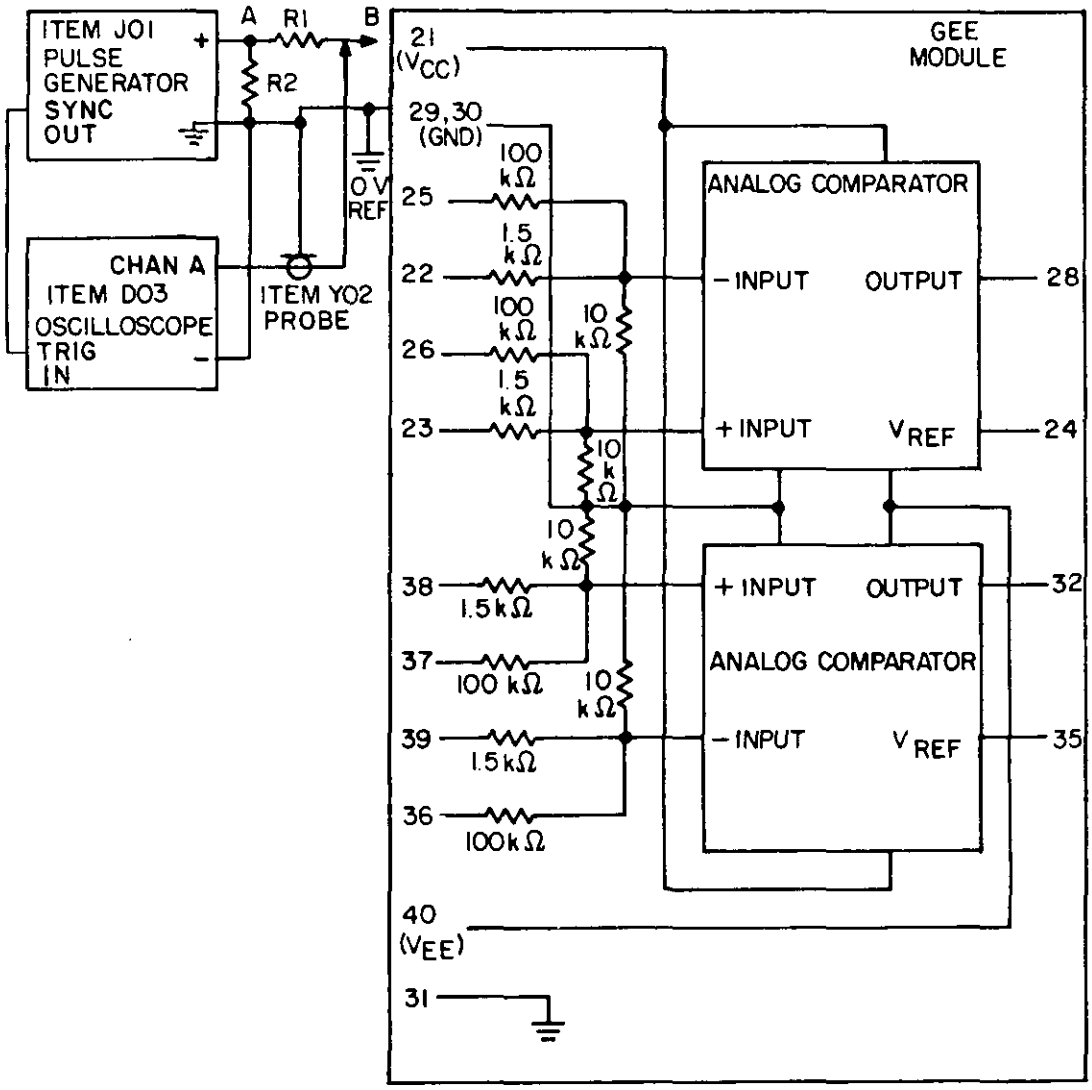
*FIGURE 21. Output saturation voltage test circuit for table XIV.

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*TABLE XV. Power supply transient rise time test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		B to pin	Min	Max		
P1	Power supply transient rise time	21	2.5	7.5	µs	22,23
P2	Power supply transient rise time	40	2.5	7.5	µs	22,23

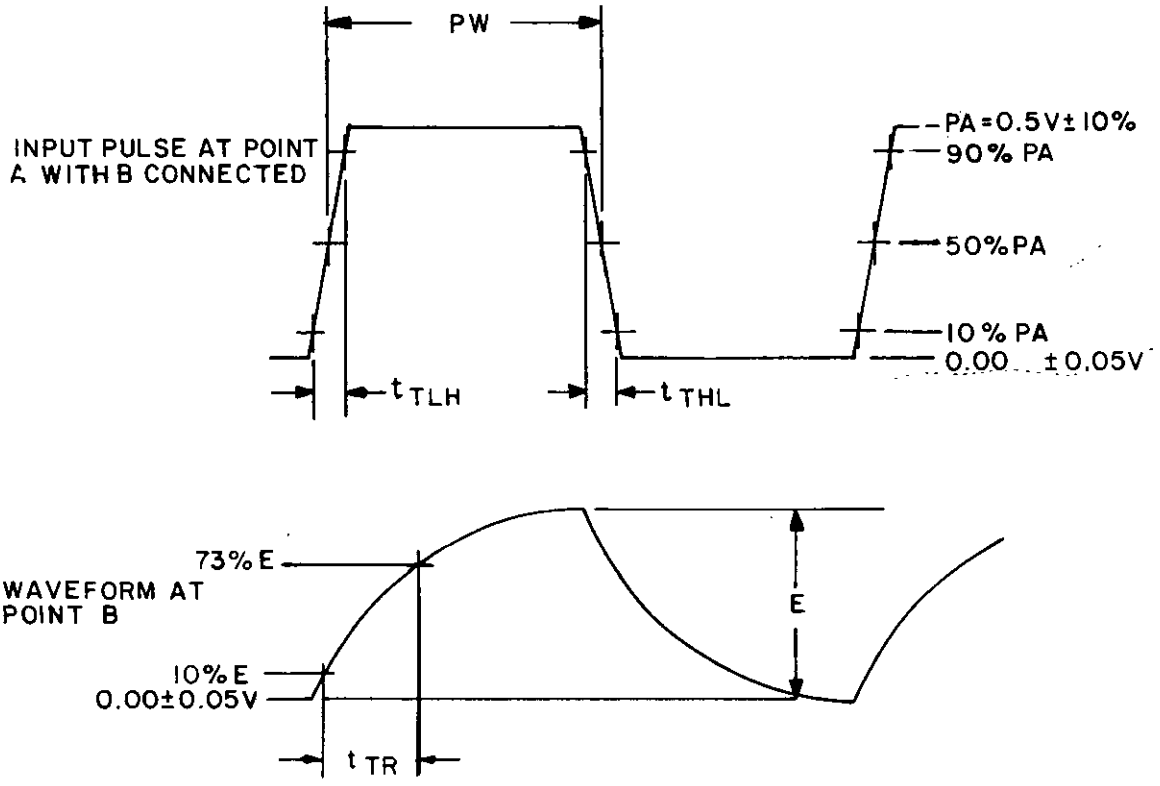
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NOTES:
1. R1 = 412Ω ±1%, 1/8 W.
2. R2 = 50Ω ±1%.

*FIGURE 22. Power supply transient rise time test circuit for table XV.

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NOTES:
1. Input pulse parameters:
   PW = 30 µs ±10%.
   PRP = 60 µs ±10%.
   $t_{TLH} = t_{THL}$ = 50 ns maximum.
2. Measure transient rise time (t_{TR}) at point B.

*FIGURE 23. Power supply transient rise time test circuit waveforms for table XV.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-28787/231A(EC)	Modules, Standard Electronic Comparator, Analog, Key Code GEE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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